Does a linear combination of eigenvectors give another eigenvector of the same matrix?

Answer:

It all depends on the eigenvalues associated with the eigenvectors you are combining.

Case 1: Eigenvectors with the Same Eigenvalue

Yes. Any linear combination of eigenvectors that share the *same* eigenvalue will also be an eigenvector with that same eigenvalue (as long as the combination does not result in the zero vector).

Let's see why. Suppose v_1 and v_2 are eigenvectors of a matrix **A**, both corresponding to the same eigenvalue r. By definition, this means:

$\mathbf{A}\mathbf{v}_1 = \mathrm{r}\mathbf{v}_1, \ \mathbf{A}\mathbf{v}_2 = \mathrm{r}\mathbf{v}_2$

Now, consider a linear combination $\mathbf{w} = \mathrm{c}_1\mathbf{v}_1 + \mathrm{c}_2\mathbf{v}_2$, where c_1 and c_2 are scalars. To check if **w** is an eigenvector, we multiply it by **A**:

$$\mathbf{A}\mathbf{w} = \mathbf{A}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2)$$

Using the linearity of matrix multiplication:

$$\mathbf{A}\mathbf{w} = c_1(\mathbf{A}\mathbf{v}_1) + c_2(\mathbf{A}\mathbf{v}_2)$$

Now, substitute the eigenvector definitions from above:

$$\mathbf{A}\mathbf{w} = c_1(r\mathbf{v}_1) + c_2(r\mathbf{v}_2)$$

Factor out the scalar eigenvalue r:

$$\mathbf{A}\mathbf{w} = r(c_1\mathbf{v}_1 + c_2\mathbf{v}_2)$$

Since $\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, we have:

$$\mathbf{A}\mathbf{w} = r\mathbf{w}$$

This is precisely the definition of an eigenvector. So, **w** is an eigenvector of **A** with the same eigenvalue r.

This is the principle behind **eigenspaces**. For a given eigenvalue r, the set of all its corresponding eigenvectors, plus the zero vector, forms a vector subspace called the eigenspace E_r. A key property of a subspace is that it is closed under linear combinations, which is exactly what we just proved.

Case 2: Eigenvectors with Different Eigenvalues

No. In general, a linear combination of eigenvectors that correspond to *different* eigenvalues is **not** an eigenvector.

Let's see why. Suppose $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of **A** with different eigenvalues, r_1 and r_2 respectively ($r_1 \neq r_2$).

$$\mathbf{A}\mathbf{v}_1 = r_1\mathbf{v}_1, \ \mathbf{A}\mathbf{v}_2 = r_2\mathbf{v}_2$$

Again, consider the linear combination $\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ (assuming c_1, c_2 are non-zero). Let's multiply by **A**:

$$\mathbf{A}\mathbf{w} = \mathbf{A}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = c_1(\mathbf{A}\mathbf{v}_1) + c_2(\mathbf{A}\mathbf{v}_2)$$

Substituting the definitions:

$$\mathbf{A}\mathbf{w} = c_1 r_1\mathbf{v}_1 + c_2 r_2\mathbf{v}_2$$

Now, for w to be an eigenvector, there would have to be a *single* scalar r_3 such that **Aw** = r_3 **w**. Let's see if this is possible:

$$c_1 r_1 \mathbf{v_1} + c_2 r_2 \mathbf{v_2} = r_3(c_1 \mathbf{v_1} + c_2 \mathbf{v_2})$$
$$c_1 r_1 \mathbf{v_1} + c_2 r_2 \mathbf{v_2} = c_1 r_3 \mathbf{v_1} + c_2 r_3 \mathbf{v_2}$$

Rearranging the terms gives:

$$(c_1 r_1 - c_1 r_3)\mathbf{v_1} + (c_2 r_2 - c_2 r_3)\mathbf{v_2} = 0$$
$$c_1(r_1 - r_3)\mathbf{v_1} + c_2(r_2 - r_3)\mathbf{v_2} = 0$$

Now eigenvectors corresponding to distinct eigenvalues are linearly independent. Since $\mathbf{v_1}$ and $\mathbf{v_2}$ are linearly independent, the only way their linear combination can be the zero vector is if their coefficients are zero.

$$c_1(r_1 - r_3) = 0, \quad c_2(r_2 - r_3) = 0$$

Since we assumed c_1 and c_2 are non-zero, this implies:

$$r_1 - r_3 = 0 \implies r_3 = r_1 \text{ and } r_2 - r_3 = 0 \implies r_3 = r_2$$

This would mean $r_1 = r_2$, which contradicts our initial assumption that the eigenvalues are different. Therefore, our premise that w could be an eigenvector must be false.

Summary

- **Same Eigenvalue:** Yes. A linear combination of eigenvectors from the same eigenspace produces another eigenvector in that same eigenspace.
- **Different Eigenvalues:** No. A linear combination of eigenvectors with different eigenvalues does not produce an eigenvector.

Scenario 1

Consider the 4x4 matrix $\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}$

Since the matrix is diagonal, the eigenvalues are the diagonal entries:

eigenvalues r = 3 and r = 5, each with multiplicity 2.

Finding Eigenvalues and Eigenvectors

1. For Eigenvalue r_1 = 3:

We need to solve the system (3**I** - **A**)**v** = **0** to get the general solution

$$\begin{pmatrix} s \\ t \\ 0 \\ 0 \end{pmatrix} = s\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

The eigenspace is spanned by the two linearly independent eigenvectors:

$$\mathbf{v_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \text{and } \mathbf{v_2} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

2. For Eigenvalue r_2 = 5:

We need to solve the system (5**I** - **A**)**v** = **0** to get the general solution

$$\begin{pmatrix} 0 \\ 0 \\ s \\ t \end{pmatrix} = s\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

The eigenspace is spanned by the two linearly independent eigenvectors:

$$\mathbf{v_3} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \text{and } \mathbf{v_4} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Verification 1: Linear Combination of Eigenvectors with the Same Eigenvalue

Let's take a linear combination of $\mathbf{v_1}$ and $\mathbf{v_2}$, which both correspond to the eigenvalue r_1 = 3. Let the combination be **w** = 2$\mathbf{v_1}$ + 4$\mathbf{v_2}$.

$$\mathbf{w} = 2\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + 4\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 0 \\ 0 \end{pmatrix}$$

Now, we must check if **w** is an eigenvector of **A** with eigenvalue 3. We do this by testing if **Aw** = 3**w**.

$$\mathbf{Aw} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}\begin{pmatrix} 2 \\ 4 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \\ 0 \\ 0 \end{pmatrix}$$

And let's calculate 3**w**: $3\mathbf{w} = 3\begin{pmatrix} 2 \\ 4 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \\ 0 \\ 0 \end{pmatrix}$

Since **Aw** = 3**w**, the vector **w** is indeed an eigenvector with eigenvalue 3. This verifies that the set of eigenvectors for a given eigenvalue (the eigenspace) is closed under linear combinations.

Verification 2: Linear Combination of Eigenvectors with Different Eigenvalues

Let's take a linear combination of $\mathbf{v_1}$ (eigenvalue 3) and $\mathbf{v_3}$ (eigenvalue 5). Let the combination be **z = $\mathbf{v_1}$ + $\mathbf{v_3}$.**

$$\mathbf{z} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Now, let's check if **z** is an eigenvector of **A**. We calculate **Az** and see if it is a scalar multiple of **z**.

$$\mathbf{Az} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 5 \\ 0 \end{pmatrix}$$

So **Az** is not a scalar multiple of **z.** Therefore, **z** is **not** an eigenvector of **A**. This verifies that a linear combination of eigenvectors with different eigenvalues is not, in general, an eigenvector.